Customer Churn & Retention Analytics: Detailed Report

Oct 15, 2026 · @anupam

Executive summary

25% of the 10,000 customers have churned (2,500 lost), and almost all of that churn sits on monthly contracts. Monthly plans churn at 44.8%, against 0.5% for quarterly and 0% for annual. Together, churned customers represent 809,074 in monthly revenue and 1.69M in lifetime value (CLTV).

- **Contract type is the dominant driver.** Monthly customers are 55.5% of the base but 99.5% of all churn (2,488 of 2,500).
- **Plan tier and price matter.** Basic churns at 34.6%, Standard at 20.1% and Premium at 12.7%. The cheapest quartile of customers churns at 34.1%, the most expensive at 16.0%.
- **Support friction is a clear signal.** Customers with an escalation churn at 33.7% versus 20.0% without. Customers who never complained churn at 8.1% versus 27.9% for those who did.
- **Price is the top stated reason**, behind 25% of cancellations and 199,782 of lost monthly revenue.
- **The model flags 229 active customers as high-risk**, carrying 74,427 in monthly revenue and 153,564 in CLTV. Nearly all are Basic, monthly subscribers.

Two cautions apply before this goes to stakeholders: the model's near-perfect scores (AUC 0.99) are inflated by a leakage issue, and the data has a September 2026 cancellation spike that looks like an artefact. Both are explained in the last section, with fixes.

Scope, data and methodology

The project analyses subscription churn for an Indian OTT streaming business: who leaves, why, what it costs, and who is likely to leave next. It runs in Python (pandas, scikit-learn), SQL on SQLite and a Looker Studio dashboard.

Dataset	Rows	Contents
Customers	10,000	ID, name, state, gender, date of birth, interests, pincode

Dataset	Rows	Contents
Subscriptions	10,000	Start/renewal/cancellation dates, subscription type, plan, contract, monthly charges, CLTV, churn score, cancellation reason
Support	19,584	Complaint date, escalation flag, CSAT score, complaint category, comment (8,550 customers)

Pipeline

1. Cleaned column names, converted dates and checked nulls and duplicates (no duplicates; only cancellation fields were null, for the 7,500 active customers).
2. Defined churn as having a cancellation date: 2,500 churned, 7,500 active.
3. Aggregated support records per customer (complaints, escalations, average CSAT) and joined all three tables on customer ID into one 10,000-row table.
4. Customers with no support record got 0 complaints and 0 escalations; their missing CSAT was filled with the median (3.5).
5. Engineered age, tenure, annual revenue, complaint and escalation flags, and risk bands (churn score 70+ high, 40-69 medium, below 40 low).
6. Produced KPIs, segment analysis, charts and matching SQL queries.
7. Trained logistic regression and random forest models (80/20 stratified split, 17 features, class-balanced) and scored every customer.

Headline KPIs

One in four customers is gone, and each lost customer cost roughly 324 per month on average (809,074 / 2,500).

KPI	Value	Note
Total customers	10,000	
Active / churned	7,500 / 2,500	
Churn rate / retention	25.0% / 75.0%	Matches the dashboard
ARPU (active)	383.97	Monthly charge per active customer
Total monthly revenue	3,688,858	All customers, including churned
Monthly revenue lost	809,074	Sum of churned customers' monthly charges

KPI	Value	Note
CLTV lost	1,688,920	Sum of churned customers' CLTV
Average tenure	1,007 days	Distorted by the analysis date, see caveats
Escalation rate	36.7%	Customers with at least one escalation
Average complaints	1.96	Per customer, including zero-complaint customers
Revenue at risk (score above 70)	75,899	Active customers; see caveats on the alternate figures

What drives churn

Contract length explains most of the churn, with plan tier and price adding a second layer.

By contract type

Contract	Customers	Churned	Churn rate
Monthly	5,552	2,488	44.8%
Quarterly	2,378	12	0.5%
Annual	2,070	0	0.0%

By plan

Plan	Customers	Churned	Churn rate
Basic	4,304	1,490	34.6%
Standard	3,859	776	20.1%
Premium	1,837	234	12.7%

By monthly price quartile

Quartile	Average charge	Avg CLTV	Churn rate
Low	195	550	34.1%
Medium-low	303	876	28.5%
Medium-high	381	1,128	21.2%

Quartile	Average charge	Avg CLTV	Churn rate
High	598	1,790	16.0%

Cheaper customers leave more, which is the opposite of what "price too high" alone would suggest. The pattern fits a low-commitment, low-engagement segment (Basic, monthly) rather than price sensitivity at the top end.

Subscription type and geography

- Family Bundle churns at 27.5%, versus 24.8% for OTT Video + Music and 24.5% for OTT Video. The gap is small.
- State churn runs from 21.5% (Madhya Pradesh) to 30.8% (Kerala). Kerala, Rajasthan, West Bengal and Tamil Nadu are above 27%. Maharashtra and Karnataka are the largest bases (1,617 and 1,232 customers) at 23.5% and 23.6%.

Tenure

The notebook shows 100% churn for customers under 6 months and 99.8% for 6-12 months, falling to 0.3% beyond 3 years. This looks dramatic but is partly an artefact of how tenure is calculated (see caveats), so treat it as directional until fixed.

Support experience and cancellation reasons

Churned customers had more complaints, more escalations and lower satisfaction than active ones, and the gap is widest between customers who contacted support and those who never did.

Measure	Active	Churned
Average complaints	1.80	2.43
Average escalations	0.41	0.66
Average CSAT (1-5)	3.56	3.31

Segment	Customers	Churn rate
No complaints	1,450	8.1%
At least one complaint	8,550	27.9%
No escalation	6,327	20.0%
Escalated	3,673	33.7%

Note that 85.5% of customers have filed a complaint, so "complaint" is a weak filter on its own; escalation is the sharper signal.

Why customers cancel (2,500 churned)

Reason	Customers	Share	Monthly revenue lost
Price too high	624	25.0%	199,782
Content dissatisfaction	398	15.9%	133,452
Not using service	343	13.7%	113,044
Competitor switch	324	13.0%	103,079
Technical issues	263	10.5%	85,793
Poor customer support	229	9.2%	73,311
Payment issues	162	6.5%	49,669
Temporary pause	157	6.3%	50,944

Roughly 40% of churn (content, non-use, competitor) is about product value and engagement. About 26% (technical, support, payment) is operational and fixable with service improvements. Temporary pauses (6.3%) are the easiest win-back group.